03 — Lifecycle Manager: issues and how we fixed them

Date: 2026-10-25 **Rule:** say what the issue was, where it lived, what changed, how we proved it. NVIDIA only unless noted.

Completeness gap (why LCM was “not done”)

LCM detect/observe/claim is live on dh2 for the NVIDIA Helm pin. Production NGC install/upgrade/uninstall of `dpf-operator` is **still not complete** on purpose (`ClusterHelmClient` refuses the live name). That is a product limit, not an unfinished detect path.

The five LCM gaps we closed in code:

Issue 1 — Helm was nil in `cmd/main.go`

Where: `cmd/main.go` **What broke:** EnsureInstalled never saw Helm storage. Live DPF looked like “detect only” even after a Secret client existed. **Fix:** wire `lifecycle.NewClusterHelmClient(mgr.GetClient())`. Empty `ChartRoot` → observe/claim only. Live name refused for mutation. **Test:** `TestMainWiresClusterHelmClient`, `TestEnsureInstalledSecretHelmReportsUnmanaged`.

Issue 2 — Live lab DPF was unmanaged

Where: Helm Secret `sh.helm.release.v1.dpf-operator.v1`, `owner=helm` **What broke:** matching pin stayed unmanaged; operators could not claim without a silent takeover. **Fix:** annotation `lifecycle.opi.nvidia.com/adopt-existing=true|pin` plus exact pin match. Stamps adapter owner + `mode=adopted`. Does not upgrade or uninstall. Cleanup default **retain**. **Lab:** annotation set; reconcile `action=noop installed=v26.4.0`. **Test:** `TestEnsureInstalledSecretHelmClaimsMatchingPin`.

Issue 3 — No OLM path

Where: installer chain **What broke:** no observation of OperatorHub; people assumed DPF was an OLM package. Latest NVIDIA docs: Helm only. **Fix:** `ClusterOLMClient` observe-only. Pin `dpf.olm.published: false`. Missing APIs / unpublished / mutation-not-configured fall through to detect. **Test:** `pkg/lifecycle/olm_client_test.go` (compiled-lab PASS on dh2).

Issue 4 — AMD has no chart

Where: `DefaultInstallConfig(VendorAMD)`, Helm/OLM skip **What broke:** skip returned **success**, hiding “no DSC operator”. **Fix:** `ErrNoVendorChart` fallthrough to `DetectingInstaller`. Do not pin AMD Network Operator or Lago. Do **not** apply `dscdevices.dpu.amd.com` stubs on BlueField (fake-device risk). **Why dh2 still NotFound:** BlueField has no Pensando (`1dd8`). NotFound is required on this host. **Test:** `amd_lcm_test.go`.

Issue 5 — Helm mutation vs k8s v0.36

Where: cannot import `helm.sh/helm/v3` **What broke:** no real install/upgrade/rollback in CI. **Fix:** `ClusterHelmClient` local charts `pkg/lifecycle/testdata/lcm-probe/{0.1.0,0.2.0}`. Dedicated release `opi-lcm-probe`. Kind spec HelmProbe **PASS**. Live `dpf-operator` still refused.

Pipeline / lab issues

Missing Argo / cert-manager looked like success

Fix: `CheckDependencies` + `SkewBlocked`. ReadyClusterProbe includes cert-manager + Argo (lab Argo lives in `dpf-operator-system`). **Test:** dependency_checker tests; lab: Argo API present.

Secret watch forbidden after first image roll

Where: `lcm-20261024` pod **What:** `secrets is forbidden: cannot watch` **Fix:** `DisableFor` Secret; roll `lcm-20261024b`. Watch errors **0**.

Live ClusterRole older than generated

What: missing dpuclusters / dpfoperatorconfigs list, config finalizer update. **Fix:** apply generated role under live name `opi-nvidia-dpf-adapter-manager-role`. No wildcards. **Test:** `TestManagerRoleHasNoWildcards`.

dh2 compiled tests FAIL without sources

What: `go test -c` binary + missing `go.mod` / `role.yaml` / `*.go` for source-read tests. **Fix:** copy those into `/tmp/lab-job-*`. Then PASS.

Kind e2e skipped (docker.sock)

Fix: `make docker-preflight` before `setup-test-e2e`. Later **6/6**.

Pin 26.4.1 temptation

Finding: official 26.4.1 is a BF3-tested Helm patch (Aug 2026). Lab is BF2, DPU already Rebooting. **Decision:** pin stays **v26.4.0**. Documented in `docs/lcm-version-pin.md`.

Scale pointers (2000-user test)

Pointer 1 — same DpuOperatorConfig 409

Error: `object was modified` on finalizer Update and Status().Update **Fix:** lock + RetryOnConflict + one worker + leader-elect true. **Proof:** 32-way unit PASS; stress 2000 same object OK.

Pointer 2 — ClusterHelm Install race

Error: `configmaps "opi-lcm-probe": object was modified` **Fix:** release lock; applyOne/storeSecret/markSuperseded retry. **Proof:** 32-way install one winner; 32-way upgrade 0.2.0; stress OK.

Pointer 3 — ClaimRelease 409

Error: `secrets "sh.helm.release.v1.dpf-operator.v1": object was modified` **Deadlock:** first ClaimRelease used the same mutex as Helm Ensure → `TestEnsureInstalledSecretHelmClaimsMatchingPin` timed out **6m**. **Fix:** `claimReleaseLocks` + retry + idempotent skip. **Proof:** second claim 0 extra Updates; 2000-way OK.

Pointer 4 — DPU Rebooting (user flow)

What: LCM Ready, dataplane not. Phase Rebooting, `host-power-cycle-required`, `rebootStatus.method=SystemLevelReset`, pending NVCONFIG PF_SF_BAR_SIZE / PF_TOTAL_SF. **Fix (software only):** `ObserveNVIDIADPUs`, `VendorDPUReady=False`, requeue 30s, message names the DPU. **No** mlxfwreset, **no** host reboot, **no** DPU spec edit. **Not fixed:** the BlueField is still Rebooting until a human does the host-agent reset.

Condition writer model (do not regress)

Object	Type	Writer
DpuOperatorConfig	Ready	vsp-opi-daemon
DpuOperatorConfig	VendorOperatorSupported	lifecycle
DpuOperatorConfig	VendorOperatorReady	lifecycle
DpuOperatorConfig	VendorOperatorCleanup	lifecycle
DpuOperatorConfig	VendorDPUReady	lifecycle
DataProcessingUnit	Ready	vsp-opi-daemon
DataProcessingUnit	DPFReady	mapping:dataprocessingunit

LCM must never write `Ready`. Tests fail if it does.

What we did not change on the server

No package install, no `/etc`, no `systemctl`, no reboot, no firmware, no Ansible, no new SSH keys, no BMC, no helm-uninstall, no AMD stubs.

How to re-prove the issue list

```
export GOTOOLCHAIN=go1.26.10
go test ./pkg/lifecycle/ -count=1
go test -race ./pkg/lifecycle/ -count=1 \
  -run 'TestConcurrentReconcile|TestConcurrentClusterHelm|TestConcurrentClaim|TestObserveNVIDIA'
make govulncheck
make docker-preflight && make test-e2e
```